

Direct Sparse Initialization for Stereo Visual-Inertial Odometry

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Abstract—Existing stereo visual-inertial initialization methods, whether tightly or loosely coupled, rely critically on intermediate variables like feature correspondences and camera poses rather than original image data. Computing these variables through feature tracking and Structure-from-Motion (SfM) inherently introduces errors, adversely affecting results. To overcome this limitation, we propose a direct initialization method for stereo visual-inertial odometry, which directly bridges original image intensities and initial parameters, bypassing conventional intermediate variables. Specifically, we introduce a prediction function to compute the corresponding points from the initial parameters. Then we formulate an objective function that optimizes initial parameters by minimizing the photometric error of sparse points, eliminating the need for feature tracking and SfM. The metric scale in our initialization is directly determined by the stereo baseline. We further propose an approximation method for two-frame initialization, demonstrating its efficacy even with minimal frame data. Extensive experiments confirm that our method achieves superior performance in both estimation accuracy and initialization success rate with shorter runtime. Even with 3 frames for initialization, our method outperforms the state-of-the-art methods using 10 frames in most metrics. Code is available at <https://github.com/Qiu0336/DSInit>.

Index Terms—Visual-inertial SLAM, sensor fusion

I. INTRODUCTION

STEREO Visual-Inertial Odometry (VIO) leverages the complementary data from a stereo camera and an Inertial Measurement Unit (IMU) to estimate a platform's egomotion. The visual stream offers rich environmental observations, whereas the IMU provides proprioceptive measurements of acceleration and angular velocity, ensuring robustness in challenging conditions such as rapid motion and texture-less scenes. Furthermore, unlike its monocular counterpart, stereo VIO directly provides metric scale through the fixed stereo baseline, enabling reliable 3D point reconstruction via triangulation. This inherent access to metric scale, combined with the low cost, small size, and minimal power consumption of the sensor suite, makes stereo VIO a highly accurate, robust, and practical solution for robotic perception and AR/VR applications.

To effectively run a stereo VIO system, some initial parameters, including initial velocity, gravity direction, gyroscope

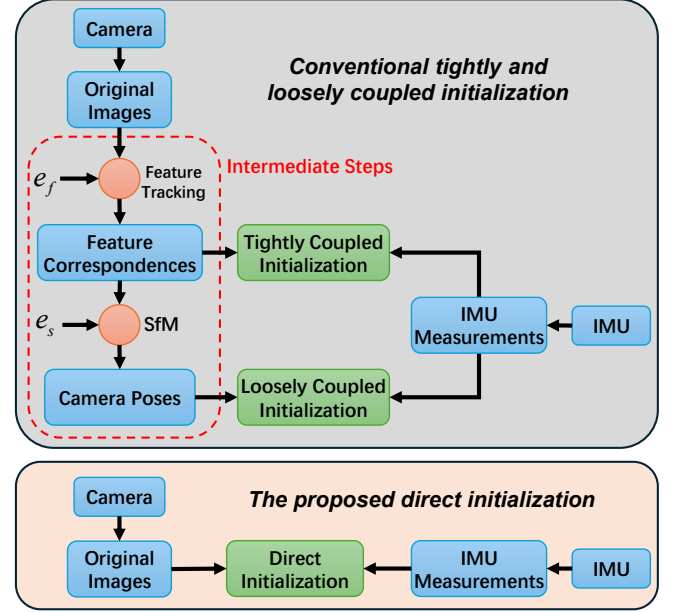


Fig. 1. Conventional tightly and loosely coupled methods require computing feature correspondences or camera poses before initialization, as indicated by the red dashed block. This computational chain can introduce additional errors (e_f and e_s), degrading performance. The proposed initialization method directly establishes constraints between original image intensities and IMU measurements, eliminating intermediate steps.

and accelerometer biases, must be accurately estimated during initialization. This process is critical, because inaccurate initial values degrade estimation accuracy and can even cause system failure. Existing stereo VIO initialization methods are either tightly or loosely coupled. Tightly coupled methods [1]–[5] derive closed-form solutions by jointly relating visual features and IMU preintegrations. However, the solutions are highly sensitive to inaccuracies in the sensory data, making them prone to numerical instability and, consequently, reduced robustness. Loosely coupled methods [6]–[10] estimate initial parameters by first solving a visual Structure-from-Motion (SfM) problem to obtain camera poses from feature correspondences, and then aligning the IMU motion with these camera poses. However, the performance of these methods is strongly tied to visual SfM, which often becomes unstable during fast motion or in scenes with too few common features.

From another perspective, both tightly and loosely coupled methods rely on a multi-stage process rather than directly constraining image intensities using IMU measurements, as depicted in Fig. 1. They first solve for intermediate variables, including feature correspondences and camera poses, and then

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use these variables to formulate the constraints and estimate initial parameters. However, establishing feature correspondences inevitably introduces errors due to feature tracking or matching, and in loosely coupled methods, additional inaccuracies arise when estimating camera poses from these correspondences. The presence of these errors, even if small, will lead to a shift in the distribution of optimal solutions. In other words, since initial parameters are estimated from intermediate variables rather than the original images, errors introduced by feature tracking or SfM in the multi-stage process cannot be corrected during initialization, resulting in biased or unreliable results.

To overcome the above limitation, we propose a direct initialization method. It adapts direct methods to the initialization stage, establishes a direct mapping from IMU initialization parameters to photometric residuals, thereby avoiding the intermediate steps that can introduce additional sources of error, as presented in Fig. 1. To the best of our knowledge, this paper represents the first attempt to adapt direct methods specifically for VIO initialization. Specifically, we first introduce a function that enables direct prediction of corresponding points from initial parameters. Based on this prediction function, we formulate an objective function to optimize the initial parameters by directly minimizing the photometric error of sparse points, bypassing intermediate steps such as feature tracking and pose estimation. We also propose an approximation method for two frame initialization to investigate the capability of our direct method in the two frame case. Exhaustive experiments demonstrate that the proposed method achieves superior accuracy and a higher success rate within a shorter runtime.

Overall, the proposed direct initialization method offers three primary advantages: 1) Our method achieves higher accuracy than existing methods because the solution distribution is dictated directly by the original images without introduced errors or shifts; 2) Our method does not require stable feature correspondences or visual initialization, resulting in a higher success rate; and 3) Our method enables faster computation by eliminating intermediate steps such as feature tracking and visual SfM.

Our contributions are highlighted in threefold:

- A prediction function is presented to connect IMU parameters with image pixels, eliminating explicit feature correspondences and camera poses.
- A novel stereo visual-inertial initialization method is proposed, which directly constrains image intensities and initial parameters, bypassing conventional steps like feature tracking and visual SfM.
- Exhaustive experiments demonstrate that the proposed initialization method achieves superior accuracy and a higher success rate within a shorter runtime.

II. RELATED WORK

Research dedicated specifically to stereo VIO initialization is relatively limited, largely because monocular VIO frameworks can be readily extended to the stereo setting. Consequently, in the following review of related work, we do

not draw an explicit distinction between monocular and stereo methods.

The tightly coupled method was pioneered by Martinelli [1], who proposed a closed-form solution constructed from a large linear system that combines visual features and IMU data. Kaiser et al. [2] improved this approach by considering gyroscope bias and constructing a cost function for refinement. Campos et al. [3] further enhanced the method by accounting for gravity magnitude and introducing two rounds of bundle adjustment. Evangelidis and Micusik [4] proposed marginalizing point positions from the initial parameters to reduce computation time. Liu et al. [5] introduced a tightly coupled method that integrates both point and line features to improve the accuracy and stability.

The loosely coupled method was first introduced by Mur-Artal and Tardós [6], in which both the gravity magnitude and IMU biases are taken into account. However, the initial parameters require more than 10 seconds to converge. Qin et al. [7] proposed a similar method but ignored accelerometer bias. They verified that the accelerometer bias has minimal effect on the results but significantly reduces the convergence time. Campos et al. [8] formulated initialization as a maximum-a-posteriori problem, where all parameters could be solved in one step. Zúñiga-Noël et al. [9] presented an analytical solution for the loosely coupled method, dramatically reducing the solving time. Wang et al. [10] incorporated an Error-state Kalman Filter to estimate gyroscope bias, overcoming dependence on pure visual SfM for rotation estimation.

Recently, He et al. [11] proposed to estimate gyroscope bias through normal epipolar constraints, and then presented a rotation-translation-decoupled framework for initialization. They implemented this framework on both tightly and loosely coupled methods. Similarly, Wang et al. [12] enhanced stereo visual inertial initialization with normal epipolar constraints, and proposed a critic to assess the initialization success in ORB-SLAM3 [13]. Mu et al. [14] extended the rotation-translation-decoupled framework by adding new uncertainty parameters and optimization modules. While these methods enable more efficient estimation of gyroscope bias, they are still dependent on feature correspondences and require feature tracking as an intermediate step, which introduces additional errors.

III. PRELIMINARIES

A. Camera Projection Model

In this paper, all images are undistorted as a pre-processing step, without accounting for camera distortion parameters. All camera frames refer to the left camera by default. The right camera is used exclusively for stereo triangulation to compute depth and is not involved in the optimization process.

We denote the coordinate of a 3D point $\mathbf{P}$ in the camera coordinate system of frame $\mathbf{I}_i$ as $\mathbf{P}_i = [X_i, Y_i, Z_i]^\top$. The equation of projection onto the image plane can be obtained using the pinhole camera model:

$$\mathbf{p}_i = \pi(\mathbf{P}_i) = \frac{1}{Z_i} \mathbf{K} \mathbf{P}_i \quad (1)$$

where $\mathbf{p}_i = [u_i, v_i]$ is the pixel coordinate on the image plane. For simplicity, homogeneous and non-homogeneous coordinates for $\mathbf{p}_i$ are used interchangeably in this paper. $\pi(\cdot)$ is the projection function, and $\mathbf{K}$ is the camera intrinsic matrix.

By using (1), the back projection equation is obtained as

$$\mathbf{P}_i = \pi^{-1}(\mathbf{p}_i, Z_i) = Z_i \mathbf{K}^{-1} \mathbf{p}_i \quad (2)$$

where the pixel coordinate $\mathbf{p}_i$ and its depth Z_i are required in the back projection function $\pi^{-1}(\cdot)$ to recover $\mathbf{P}_i$.

B. IMU Preintegration

IMU preintegration aims to obtain the relative motion increments that are independent of the absolute pose and velocity by integrating IMU measurements over a specific time interval. According to the theory on manifold [15], the discrete state increments from time i to j are expressed as

$$\mathbf{R}_{wbj} = \mathbf{R}_{wbi} \Delta \tilde{\mathbf{R}}_{ij} \quad (3)$$

$$\mathbf{v}_{bj}^w = \mathbf{v}_{bi}^w + \mathbf{g}^w \Delta t_{ij} + \mathbf{R}_{wbi} \Delta \tilde{\mathbf{v}}_{ij} \quad (4)$$

$$\mathbf{p}_{wbj} = \mathbf{p}_{wbi} + \mathbf{v}_{bi}^w \Delta t_{ij} + \frac{1}{2} \mathbf{g}^w \Delta t_{ij}^2 + \mathbf{R}_{wbi} \Delta \tilde{\mathbf{p}}_{ij} \quad (5)$$

where $\mathbf{R}_{wbi}$ and $\mathbf{p}_{wbi}$ are the orientation and position of IMU at time i in the world coordinate system. Δt_{ij} is the time interval from i to j . $\mathbf{v}_{bi}^w$ and $\mathbf{g}^w$ represent IMU velocity at time i and gravity vector in the world coordinate system. In this paper, we define the positive Z -axis as pointing upward, therefore the expected value of $\mathbf{g}^w$ is approximately $(0, 0, -9.81)$. $\{\Delta \tilde{\mathbf{R}}_{ij}, \Delta \tilde{\mathbf{v}}_{ij}, \Delta \tilde{\mathbf{p}}_{ij}\}$ are the preintegrated measurements that can be directly computed from the IMU measured values:

$$\Delta \tilde{\mathbf{R}}_{ij} = \prod_{k=i}^{j-1} \exp((\tilde{\omega}_k - \mathbf{b}_k^g) \Delta t) \quad (6)$$

$$\Delta \tilde{\mathbf{v}}_{ij} = \sum_{k=i}^{j-1} \Delta \tilde{\mathbf{R}}_{ik} ((\tilde{\mathbf{a}}_k - \mathbf{b}_k^a) \Delta t) \quad (7)$$

$$\Delta \tilde{\mathbf{p}}_{ij} = \sum_{k=i}^{j-1} \left[\Delta \tilde{\mathbf{v}}_{ik} \Delta t + \frac{1}{2} \Delta \tilde{\mathbf{R}}_{ik} ((\tilde{\mathbf{a}}_k - \mathbf{b}_k^a) \Delta t^2) \right] \quad (8)$$

where $\exp(\cdot)$ is the vector-valued exponential map of $\text{SO}(3)$. $\tilde{\omega}_k$ and $\tilde{\mathbf{a}}_k$ indicate the gyroscope and accelerometer measurements at time k , while $\mathbf{b}_k^g$ and $\mathbf{b}_k^a$ indicate the IMU biases. Δt denotes the IMU sampling interval.

We observed that jointly optimizing gravity and accelerometer bias $\mathbf{b}^a$ during initialization can lead to simultaneous variations and convergence to unrealistic values. This demonstrates the strong coupling between gravity and $\mathbf{b}^a$ [6], and the high sensitivity of the our formulation to linear-acceleration errors. Therefore, similar to [11], we assume that $\mathbf{b}^a$ is zero in this paper, which has only a minor impact on initialization performance [7], [16]. The effect of gyroscope bias on the preintegrated measurements can be computed using first-order Taylor approximation as follows:

$$\Delta \tilde{\mathbf{R}}_{ij}(\mathbf{b}^g) \approx \Delta \tilde{\mathbf{R}}_{ij} \exp(\mathbf{J}_{\mathbf{b}^g}^{\Delta \tilde{\mathbf{R}}_{ij}} \mathbf{b}^g) \quad (9)$$

$$\Delta \tilde{\mathbf{v}}_{ij}(\mathbf{b}^g) \approx \Delta \tilde{\mathbf{v}}_{ij} + \mathbf{J}_{\mathbf{b}^g}^{\Delta \tilde{\mathbf{v}}_{ij}} \mathbf{b}^g \quad (10)$$

$$\Delta \tilde{\mathbf{p}}_{ij}(\mathbf{b}^g) \approx \Delta \tilde{\mathbf{p}}_{ij} + \mathbf{J}_{\mathbf{b}^g}^{\Delta \tilde{\mathbf{p}}_{ij}} \mathbf{b}^g \quad (11)$$

where $\mathbf{b}^g$ is the gyroscope bias to be estimated during initialization. $\{\mathbf{J}_{\mathbf{b}^g}^{\Delta \tilde{\mathbf{R}}_{ij}}, \mathbf{J}_{\mathbf{b}^g}^{\Delta \tilde{\mathbf{v}}_{ij}}, \mathbf{J}_{\mathbf{b}^g}^{\Delta \tilde{\mathbf{p}}_{ij}}\}$ represent the Jacobian of $\{\Delta \tilde{\mathbf{R}}_{ij}, \Delta \tilde{\mathbf{v}}_{ij}, \Delta \tilde{\mathbf{p}}_{ij}\}$ with respect to the gyroscope bias $\mathbf{b}^g$.

C. Camera-IMU Transformation

Assuming that the camera and IMU are extrinsically calibrated as $\mathbf{R}_{bc}$ and $\mathbf{p}_{bc}$, which relate the poses between camera frame and IMU body frame as follows:

$$\mathbf{R}_{wc} = \mathbf{R}_{wb} \mathbf{R}_{bc}, \quad \mathbf{p}_{wc} = \mathbf{R}_{wb} \mathbf{p}_{bc} + \mathbf{p}_{wb} \quad (12)$$

where $\{\mathbf{R}_{wc}, \mathbf{p}_{wc}\}$ and $\{\mathbf{R}_{wb}, \mathbf{p}_{wb}\}$ are the poses of camera and IMU in the world coordinate system.

The camera's relative rotation and translation from time i to j are denoted as $\mathbf{R}_{cjc_i}$ and $\mathbf{t}_{cjc_i}$. Based on (12), we derive the IMU's relative rotation $\mathbf{R}_{bjb_i}$ and translation $\mathbf{t}_{bjb_i}$ through

$$\begin{aligned} \mathbf{R}_{cjc_i} &= \mathbf{R}_{bc}^\top \mathbf{R}_{bjb_i} \mathbf{R}_{bc} \\ \mathbf{t}_{cjc_i} &= \mathbf{R}_{bc}^\top (\mathbf{R}_{bjb_i} \mathbf{p}_{bc} + \mathbf{t}_{bjb_i} - \mathbf{p}_{bc}) \end{aligned} \quad (13)$$

which relates the relative transformation with respect to camera frame and IMU body frame.

IV. METHODOLOGY

A. Prediction Function

Let us consider a map point $\mathbf{P}$, observed by a camera in two different frames $\{c_i, c_j\}$. By geometrically transforming the coordinates of $\mathbf{P}$ from c_i to c_j , we have

$$\mathbf{P}_{c_j} = \mathbf{R}_{cjc_i} \mathbf{P}_{c_i} + \mathbf{t}_{cjc_i} \quad (14)$$

where $\mathbf{P}_{c_i}$ and $\mathbf{P}_{c_j}$ are the coordinates of $\mathbf{P}$ with respect to camera frame c_i and c_j , respectively. For the visual and inertial system with known extrinsic parameters $\mathbf{R}_{bc}$ and $\mathbf{p}_{bc}$, substituting (13) into (14) gives a transformation in the IMU body frame as follows:

$$\mathbf{P}_{c_j} = \mathbf{R}_{bc}^\top (\mathbf{R}_{bjb_i} (\mathbf{R}_{bc} \mathbf{P}_{c_i} + \mathbf{p}_{bc}) + \mathbf{t}_{bjb_i} - \mathbf{p}_{bc}) \quad (15)$$

This transformation relates $\mathbf{P}_{c_i}$ and $\mathbf{P}_{c_j}$ through the relative transformation $\{\mathbf{R}_{bjb_i}, \mathbf{t}_{bjb_i}\}$. Note that this transformation can be derived from IMU preintegrations, initial velocity and gravity in (3) and (5). Therefore, applying them to (15) and rearranging terms yields

$$\begin{aligned} \mathbf{P}_{c_j} &= \mathbf{R}_{bc}^\top \left[\Delta \tilde{\mathbf{R}}_{ij}^\top (\mathbf{R}_{bc} \mathbf{P}_{c_i} + \mathbf{p}_{bc}) \right. \\ &\quad \left. - \Delta \tilde{\mathbf{p}}_{ij} - \mathbf{v}_{bi}^{b_i} \Delta t_{ij} - \frac{1}{2} \mathbf{g}^{b_i} \Delta t_{ij}^2 - \mathbf{p}_{bc} \right] \end{aligned} \quad (16)$$

where $\mathbf{v}_{bi}^{b_i}$ and $\mathbf{g}^{b_i}$ represent IMU velocity at time i and gravity vector in the IMU body frame b_i , which are to be estimated during initialization.

We seek to establish the relationship between pixel coordinates, i.e. $\mathbf{p}_j$ and $\mathbf{p}_i$ on the image plane, rather than between 3D points. Consequently, applying the projection function $\mathbf{p}_j = \pi(\mathbf{P}_{c_j})$ and the back projection function $\mathbf{P}_{c_i} = \pi^{-1}(\mathbf{p}_i, Z_i)$ in (16) yields

$$\begin{aligned} \mathbf{p}_j &= \pi \left[\mathbf{R}_{bc}^\top \left(\Delta \tilde{\mathbf{R}}_{ij}^\top (\mathbf{R}_{bc} \pi^{-1}(\mathbf{p}_i, Z_i) + \mathbf{p}_{bc}) \right. \right. \\ &\quad \left. \left. - \Delta \tilde{\mathbf{p}}_{ij} - \mathbf{v}_{bi}^{b_i} \Delta t_{ij} - \frac{1}{2} \mathbf{g}^{b_i} \Delta t_{ij}^2 - \mathbf{p}_{bc} \right) \right]. \end{aligned} \quad (17)$$

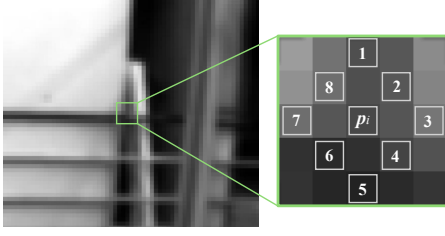


Fig. 2. Illustration of the set $\mathcal{N}_{\mathbf{p}_i}$, which comprises $\mathbf{p}_i$ along with its eight neighboring pixels. All pixels within $\mathcal{N}_{\mathbf{p}_i}$ share a common depth value Z_i .

This equation describes how to compute the corresponding point $\mathbf{p}_j$ on image $\mathbf{I}_j$ from a point $\mathbf{p}_i$ on image $\mathbf{I}_i$, with the unknown parameters comprising $\Delta\mathbf{R}_{ij}$, $\Delta\mathbf{p}_{ij}$, $\mathbf{v}_{b_i}^{b_i}$, $\mathbf{g}^{b_i}$, and Z_i . In stereo vision, the depth Z_i can be obtained through stereo correspondence followed by triangulation. Therefore, we treat Z_i as a known quantity in our formulation. $\Delta\mathbf{R}_{ij}$ and $\Delta\mathbf{p}_{ij}$ can be obtained by preintegrating IMU data, as shown in (6) and (8). However, these preintegrations are dependent on the gyroscope bias $\mathbf{b}^g$. We can treat $\Delta\mathbf{R}_{ij}$ and $\Delta\mathbf{p}_{ij}$ as functions of $\mathbf{b}^g$, effectively considering $\mathbf{b}^g$ as a new parameter that substitutes for $\Delta\mathbf{R}_{ij}$ and $\Delta\mathbf{p}_{ij}$. At this stage, the unknown parameters become $\mathbf{b}^g$, $\mathbf{v}_{b_i}^{b_i}$ and $\mathbf{g}^{b_i}$. We formalize (17) as a function $\mathbf{f}$:

$$\hat{\mathbf{p}}_j \triangleq \mathbf{f}(\mathbf{p}_i, \mathbf{b}^g, \mathbf{v}_{b_i}^{b_i}, \mathbf{g}^{b_i}). \quad (18)$$

In this paper, we refer to $\mathbf{f}$ as “prediction function”. Given a point $\mathbf{p}_i$ on the reference image $\mathbf{I}_i$, the prediction function $\mathbf{f}$ aims to compute the predicted corresponding point $\hat{\mathbf{p}}_j$ on the target image $\mathbf{I}_j$, where the unknown parameters to be estimated include $\mathbf{b}^g$, $\mathbf{v}_{b_i}^{b_i}$, and $\mathbf{g}^{b_i}$. The prediction function bridges the gap between initial parameters and pixel coordinates, paving the way for subsequent direct initialization.

B. Direct Initialization Method

For a reference image $\mathbf{I}_i$ and a target image $\mathbf{I}_j$, the image intensities at $\mathbf{p}_i$ and the predicted $\hat{\mathbf{p}}_j$ are denoted as $\mathbf{I}_i(\mathbf{p}_i)$ and $\mathbf{I}_j(\hat{\mathbf{p}}_j)$, respectively. We present the photometric residual at pixel $\mathbf{p}_i$ (i.e. the difference in intensity) as

$$\mathbf{r}_{ij} = \mathbf{I}_j(\hat{\mathbf{p}}_j) - \alpha_j^i \mathbf{I}_i(\mathbf{p}_i) + \beta_j^i \quad (19)$$

where the coefficients α_j^i and β_j^i are introduced to account for image changes in the contrast and brightness between $\mathbf{I}_i$ and $\mathbf{I}_j$. We initialize $\alpha_j^i = 1$ and $\beta_j^i = 0$.

Since $\hat{\mathbf{p}}_j$ can be predicted via the prediction function (18), substituting (18) into (19) yields

$$\mathbf{r}_{ij} = \mathbf{I}_j(\mathbf{f}(\mathbf{p}_i, \mathbf{b}^g, \mathbf{v}_{b_i}^{b_i}, \mathbf{g}^{b_i})) - \alpha_j^i \mathbf{I}_i(\mathbf{p}_i) + \beta_j^i. \quad (20)$$

This provides the photometric residual through initial parameters $\mathbf{b}^g$, $\mathbf{v}_{b_i}^{b_i}$, $\mathbf{g}^{b_i}$ at pixel $\mathbf{p}_i$.

We define the photometric error function of feature $\mathbf{p}_i$ as the sum of squared $\mathbf{r}_{ij}$ over a small neighborhood of pixels:

$$\mathbf{e}_{ij} \triangleq \mathbf{e}_{ij}(\mathbf{p}_i, \mathbf{b}^g, \mathbf{v}_{b_i}^{b_i}, \mathbf{g}^{b_i}, \alpha_j^i, \beta_j^i) = \sum_{\mathbf{p}_i \in \mathcal{N}_{\mathbf{p}_i}} \rho(\|\mathbf{r}_{ij}\|^2) \quad (21)$$

where $\rho(\cdot)$ is the Huber kernel function [17]. $\mathcal{N}_{\mathbf{p}_i}$ is a set comprising $\mathbf{p}_i$ along with its eight neighboring pixels, as

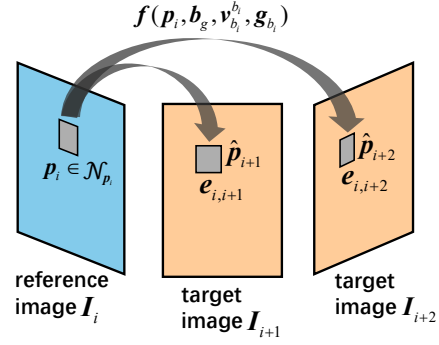


Fig. 3. Schematic diagram of photometric error generation in $N = 3$ case. The points in $\mathcal{N}_{\mathbf{p}_i}$ will generate a photometric error $\mathbf{e}_{ij}$ on each target image $\mathbf{I}_{i+1}$ and $\mathbf{I}_{i+2}$. The predicted point $\hat{\mathbf{p}}_j$ on the target frame is computed via the prediction function $\mathbf{f}$, then $\mathbf{e}_{ij}$ is generated using (21).

depicted in Fig. 2. This set definition is inspired by DSO [18], with all pixels within $\mathcal{N}_{\mathbf{p}_i}$ sharing a common depth value Z_i . The error function (21) directly links image intensity to the IMU initial parameters. This enables the direct optimization of these parameters by minimizing the photometric error, without relying on feature tracking and visual SfM. More importantly, it establishes a continuous and differentiable mapping from initial parameters to photometric residuals, enabling inherent errors to be gradually corrected through joint optimization rather than resulting in unreliable estimates typical of conventional multi-stage initialization.

Let us consider a sequence of N consecutive image frames (from i to $i+N-1$), with the first image $\mathbf{I}_i$ being the reference image, and the subsequent images serving as the target images $\mathbf{I}_j, j \in \{i+1, i+2, \dots, i+N-1\}$. According to (21), the points in $\mathcal{N}_{\mathbf{p}_i}$ will generate a photometric error $\mathbf{e}_{ij}$ on each target image $\mathbf{I}_j$, resulting in $N-1$ photometric errors, as illustrated in Fig. 3. Assuming that M sparse points are uniformly detected on $\mathbf{I}_i$, they will lead to a total of $M(N-1)$ photometric errors. Let $\mathcal{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_M\}$ denotes the set of all points on image $\mathbf{I}_i$. By aggregating all photometric errors, we obtain the final objective function:

$$\mathbf{x}^* \triangleq \underset{\mathbf{x}}{\operatorname{argmin}} \sum_{j=i+1}^{i+N-1} \sum_{\mathbf{p}_i \in \mathcal{P}} \mathbf{e}_{ij}(\mathbf{p}_i, \mathbf{b}^g, \mathbf{v}_{b_i}^{b_i}, \mathbf{g}^{b_i}, \alpha_j^i, \beta_j^i) \quad (22)$$

where the decision variable $\mathbf{x}$ contains all the unknown parameters to be estimated and is written as

$$\mathbf{x} = \{\mathbf{b}^g, \mathbf{v}_{b_i}^{b_i}, \mathbf{g}^{b_i}, \alpha, \beta\} \quad (23)$$

with the vectors $\alpha = [\alpha_{i+1}^i, \alpha_{i+2}^i, \dots, \alpha_{i+N-1}^i]$ and $\beta = [\beta_{i+1}^i, \beta_{i+2}^i, \dots, \beta_{i+N-1}^i]$. Distinct α and β are employed for different target frames, whereas all points within the same target frame share a common α and β . We provide an ablation study in Section V-E to validate the effectiveness of α and β .

At this point, the initial parameters $\{\mathbf{b}^g, \mathbf{v}_{b_i}^{b_i}, \mathbf{g}^{b_i}\}$ can be optimized by minimizing the objective function (22). Gauss-Newton algorithm is used for optimization, with the initial values $\mathbf{b}^g = \mathbf{v}_{b_i}^{b_i} = \mathbf{0}$, $\mathbf{g}^{b_i} = -\Delta\tilde{\mathbf{v}}_{i,i+1}/\Delta t_{i,i+1}$, $\alpha = \mathbf{1}$ and $\beta = \mathbf{0}$. To enhance robustness, we apply an image pyramid with a coarse-to-fine optimization strategy, where the optimization proceeds sequentially from the coarsest to

the finest level, and the estimates at each level are used to initialize the subsequent finer-level optimization. The number of pyramid levels is determined empirically through experimental evaluation, and 5 levels are adopted since performance saturates beyond this point. The gravity magnitude is utilized as an evaluation criterion to reject results with large magnitude errors. Once the initial parameters are obtained, the IMU preintegration is recomputed. Finally, the relative poses can be propagated through IMU based on (3) and (5).

C. Two-Frame Initialization

Note that the necessary condition for the initial parameters to be solvable in function (22) is $N \geq 3$. In the two-frame case when $N = 2$, the initial velocity and gravity are coupled and unobservable. This is because all photometric errors share the same Δt_{ij} , i.e. $\Delta t_{i,i+1}$, making it impossible to distinguish between $\mathbf{v}_{b_i}^{b_i}$ and $\mathbf{g}^{b_i}$ from the term $\mathbf{v}_{b_i}^{b_i} \Delta t_{ij} + \frac{1}{2} \mathbf{g}^{b_i} \Delta t_{ij}^2$ in (17). To investigate the capability of direct initialization in this limiting case, an approximation method is proposed for two-frame initialization.

Let us revisit $\mathbf{R}_{b_j b_i}$ and $\mathbf{t}_{b_j b_i}$ in (15), but now we substitute only $\mathbf{R}_{b_j b_i}$ with its preintegration while retaining $\mathbf{t}_{b_j b_i}$. After using the models (1) and (2), we obtain

$$\mathbf{p}_j = \pi \left[\mathbf{R}_{bc}^\top (\Delta \tilde{\mathbf{R}}_{ij}^\top (\mathbf{R}_{bc} \boldsymbol{\pi}^{-1}(\mathbf{p}_i, Z_i) + \mathbf{p}_{bc}) + \mathbf{t}_{b_j b_i} - \mathbf{p}_{bc}) \right] \quad (24)$$

where $j = i + 1$. $\mathbf{t}_{b_j b_i}$ is functionally dependent on $\mathbf{v}_{b_i}^{b_i}$ and $\mathbf{g}^{b_i}$, and can be treated as a new unknown parameter. Following the same derivations from (18) to (22), the set of parameters to be optimized in the objective function becomes $\mathbf{x} = \{\mathbf{b}^g, \mathbf{t}_{b_{i+1} b_i}, \boldsymbol{\alpha}, \boldsymbol{\beta}\}$, which can be well estimated in the two-frame case.

In practice, since the sampling time interval between two consecutive frames i and $i + 1$ is typically short (e.g., 0.05 seconds for a 20 Hz camera), we employ the average inter-frame velocity $\bar{\mathbf{v}}_{i,i+1}^{b_i}$ as the initial velocity $\mathbf{v}_{b_i}^{b_i}$. Given the solutions for $\mathbf{b}^g$ and $\mathbf{t}_{b_{i+1} b_i}$, the initial velocity is approximated as

$$\mathbf{v}_{b_i}^{b_i} \approx \bar{\mathbf{v}}_{i,i+1}^{b_i} = -\Delta \tilde{\mathbf{R}}_{i,i+1}(\mathbf{b}^g) \mathbf{t}_{b_{i+1} b_i} / \Delta t_{i,i+1} \quad (25)$$

and the gravity vector is approximated as

$$\mathbf{g}^{b_i} \approx -\Delta \tilde{\mathbf{v}}_{i,i+1}(\mathbf{b}^g) / \Delta t_{i,i+1}. \quad (26)$$

It is emphasized that in the two frame initialization, only $\mathbf{b}^g$ is observable, whereas $\mathbf{v}_{b_i}^{b_i}$ and $\mathbf{g}^{b_i}$ are inherently unobservable. The approximations given by (25) and (26) are reliable only when the linear acceleration of the IMU is small at time i and $i + 1$, particularly under stationary or uniform motion conditions. Large IMU accelerations may lead to significant deviations, which will be empirically validated in Section V-C.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

Experiments are conducted on both EuRoC dataset [19] and TUM-VI dataset [20] to evaluate the proposed method. For N consecutive image frames in the experiments, the first image is designated as the reference image, while the subsequent serve as the target images. Point features are detected uniformly

on the reference image, then the depth of each point is calculated through stereo matching and triangulation. When $N \geq 3$, we apply the objective function (22) for initialization, whereas when $N = 2$, the result is approximated using the method outlined in Section IV-C. The image frame rate in our experiments is maintained at 20 Hz, consistent with the native rate of the EuRoC and TUM-VI datasets. Instead of discarding any frames that might cause initialization failure (e.g. by checking the IMU excitation), we initialize all the input frames sequentially, thereby effectively evaluating the algorithm's initialization success rate. All experiments are carried out on an Intel Core Ultra 9 275HX computer with 16 GB of RAM. For comparison, we select baseline methods that explicitly address stereo visual-inertial initialization and provide publicly available implementations, enabling a fair and reproducible evaluation under comparable stereo-inertial settings. The following three stereo VIO initialization methods are considered:

- **VINS**: VINS-Fusion [21] initialization method [7], where the scale is fixed in stereo case.
- **MAP**: Inertial-only maximum-a-posteriori estimation method [8] which is implemented in ORB-SLAM3 [13].
- **SNEC**: Enhanced stereo visual inertial initialization [12] via normal epipolar constraints.

B. Three-Frame Initialization

We first conduct the experiments for the three frames setting, as $N = 3$ is the minimum number of frames when the initialization problem is solvable. Six metrics are used to evaluate the performance, including the gyroscope bias, initial velocity, gravity direction, relative translation error (RTE), relative rotation error (RRE), and success rate. The metric scale is determined by the stereo baseline for all compared methods. Note the success rate of initialization is a crucial metric and should not be ignored in favor of error metrics only, since different methods employ distinct criteria for determining the failure, with some prioritizing a higher success rate at the expense of precision.

Results on the EuRoC dataset are shown in Table. I. Our method demonstrates consistently lower errors across all metrics than the baselines in all sequences except for V103 and V203. This performance gap can be attributed to the following factors: When only a limited number of frames are available, sequences with weak texture, illumination variations, or small inter-frame motion often challenge feature tracking and SfM, leading to unstable or inaccurate initialization in baseline methods. By directly leveraging photometric information without explicit feature correspondence or geometric reconstruction, our method is less affected by these factors, which explains its superior performance under low-frame initialization settings.

Moreover, the errors of our method on V103 and V203 are noticeably larger than those on other sequences (e.g. $\mathbf{b}^g$ error of 11.47% and 32.06%, respectively, while others are below 6%; the figures for velocity are 0.045 and 0.09, while others are below 0.03). This is mainly due to the strong motion blur present in these sequences, which degrades the reliability of

TABLE I
EXHAUSTIVE INITIALIZATION RESULTS FOR THE THREE FRAMES SETTING ON THE EUROC DATASET

		MH01	MH02	MH03	MH04	MH05	V101	V102	V103	V201	V202	V203
Gyro Bias RMSE (%)	VINS	4.02	4.82	3.78	4.01	5.14	5.35	9.67	10.51	10.07	12.27	25.95
	MAP	4.60	5.90	4.90	7.02	7.42	10.34	20.33	25.99	11.81	19.96	45.11
	SNEC	3.29	3.64	2.94	3.69	4.16	5.79	6.67	7.47	10.17	11.23	11.18
	Ours	1.90	2.41	2.76	2.94	2.55	2.29	4.72	11.47	2.93	5.84	32.06
Velocity RMSE (m/s)	VINS	0.023	0.025	0.035	0.044	0.051	0.032	0.051	0.051	0.035	0.043	0.089
	MAP	0.030	0.031	0.051	0.104	0.085	0.049	0.086	0.122	0.044	0.077	0.153
	SNEC	0.027	0.028	0.030	0.050	0.051	0.038	0.049	0.068	0.041	0.053	0.061
	Ours	0.010	0.010	0.019	0.023	0.024	0.012	0.024	0.045	0.011	0.021	0.090
Gravity Dir. RMSE (deg)	VINS	3.49	3.74	4.72	4.73	5.50	5.13	6.94	7.53	4.25	5.46	11.52
	MAP	4.44	4.77	7.26	10.05	9.35	8.16	13.95	16.66	5.89	10.60	21.01
	SNEC	4.09	4.27	4.60	6.22	7.12	6.36	7.90	8.27	4.99	6.66	7.80
	Ours	1.71	1.60	2.15	2.98	2.77	3.31	2.44	2.98	1.54	2.31	6.48
Relative Rot. (RRE) (deg)	VINS	0.027	0.032	0.030	0.029	0.032	0.049	0.075	0.077	0.090	0.090	0.175
	MAP	0.032	0.038	0.040	0.048	0.046	0.080	0.138	0.172	0.102	0.133	0.274
	SNEC	0.027	0.029	0.024	0.029	0.031	0.058	0.066	0.076	0.092	0.087	0.110
	Ours	0.008	0.011	0.014	0.014	0.012	0.025	0.034	0.070	0.019	0.034	0.162
Relative Trans. (RTE) (m)	VINS	0.0018	0.0019	0.0030	0.0032	0.0039	0.0029	0.0039	0.0038	0.0023	0.0031	0.0076
	MAP	0.0023	0.0024	0.0043	0.0069	0.0065	0.0043	0.0076	0.0095	0.0030	0.0062	0.0134
	SNEC	0.0021	0.0021	0.0023	0.0034	0.0037	0.0033	0.0034	0.0039	0.0026	0.0033	0.0040
	Ours	0.0009	0.0008	0.0021	0.0023	0.0023	0.0021	0.0025	0.0040	0.0009	0.0019	0.0083
Success Rate (%)	VINS	79.2	75.8	59.7	57.9	56.0	68.1	38.1	29.5	61.9	46.2	17.7
	MAP	100.0	98.8	99.8	99.3	100.0	99.3	96.0	93.6	95.9	98.6	66.6
	SNEC	84.2	81.8	42.6	50.7	56.6	82.2	20.1	29.2	84.4	33.6	15.5
	Ours	94.6	95.3	95.6	92.2	96.6	98.9	98.3	96.9	96.9	96.0	58.8

TABLE II
3-FRAME INITIALIZATION RESULTS ON THE TUM-VI DATASET

		room1	room2	room3	room4	room5	room6
G. Dir. (deg)	VINS	5.99	5.71	5.23	4.64	5.78	4.17
	MAP	14.06	13.17	10.39	10.64	12.60	8.53
	SNEC	8.67	7.57	7.34	7.91	8.66	7.13
	Ours	2.33	2.28	1.98	2.11	2.72	1.96
RRE (deg)	VINS	0.253	0.380	0.387	0.235	0.212	0.224
	MAP	0.331	0.445	0.422	0.283	0.310	0.242
	SNEC	0.256	0.382	0.370	0.250	0.234	0.225
	Ours	0.248	0.376	0.297	0.234	0.229	0.204
RTE (m)	VINS	0.0036	0.0048	0.0039	0.0024	0.0030	0.0026
	MAP	0.0085	0.0092	0.0068	0.0064	0.0082	0.0049
	SNEC	0.0048	0.0058	0.0049	0.0042	0.0048	0.0041
	Ours	0.0034	0.0052	0.0025	0.0027	0.0030	0.0022
S. Rate (%)	VINS	37.8	39.1	53.6	47.3	37.4	55.3
	MAP	96.9	96.1	95.7	95.2	94.3	99.1
	SNEC	85.1	83.4	88.7	89.3	83.6	96.4
	Ours	98.4	99.2	88.1	97.9	97.4	100.0

photometric residuals by violating the brightness constancy assumption and attenuating image gradients. Consequently, the optimization becomes less informative and more prone to convergence issues, adversely affecting direct initialization. In comparison, methods that rely on feature correspondences are less susceptible to motion blur.

Regarding the success rate, our method maintains a rate above 90% across all sequences except for V203. In comparison, both VINS and SNEC exhibit success rates below 90% with considerable fluctuations. Although MAP achieves nearly 100% on the first six sequences, this comes at the sacrifice of precision, resulting in generally larger estimation errors

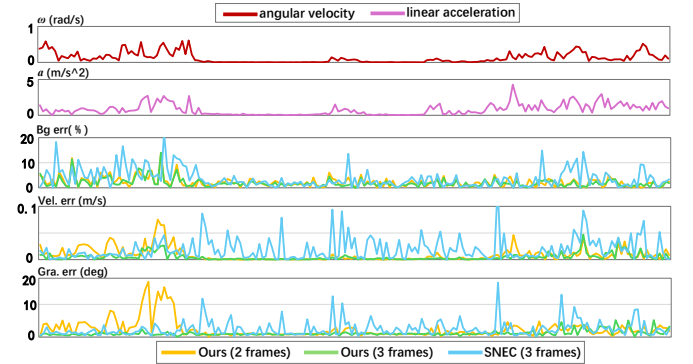


Fig. 4. Results on MH01 sequence, where the first 200 successful results are presented. The comparative methods comprise our two-frame and three-frame initialization methods, and SNEC's three-frame initialization.

compared to other methods.

An additional evaluation is performed on the TUM-VI dataset. The comparative results are presented in Table II. Notably, comparisons for the gyroscope bias and velocity are excluded because TUM-VI does not provide ground truth for them. The results demonstrate that our method achieves superior accuracy in gravity across all sequences. Furthermore, our method outperforms the benchmarks in both RRE and RTE for most sequences. While VINS achieves superior RTE in rooms 2&4 and superior RRE in room 5, its success rate is only 30-60%. Our method maintains a success rate largely above 90%, with room 3 (88.1%) being the only exception. In summary, our method yields state-of-the-art results in both the initialization accuracy and success rate.

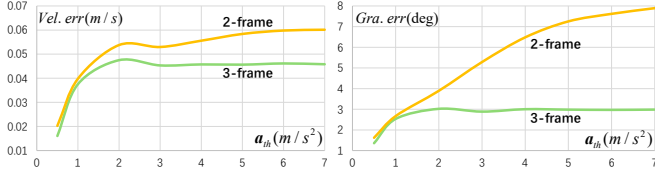


Fig. 5. Velocity and gravity errors for 2-frame and 3-frame initialization under different acceleration thresholds.

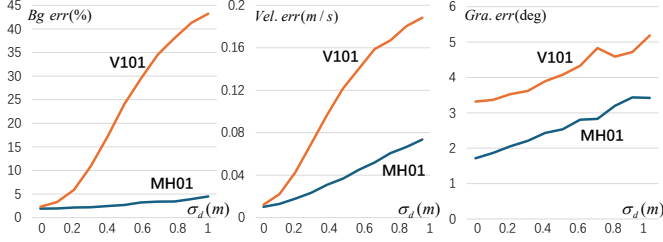


Fig. 6. Sensitivity of 3-frame initialization results to depth error: depth with Gaussian noise of standard deviation σ_d (σ_d from 0 to 1 m).

C. Two-Frame Initialization and Robustness Evaluation

We have also validated the proposed two-frame initialization method. As shown in Fig. 4, we sequentially visualized the first 200 successful initialization results on the MH01 sequence, with the comparative methods including our 2-frame and 3-frame initialization, and SNEC’s 3-frame initialization. The ground-truth angular velocity and linear acceleration are also visualized synchronously for reference. For $\mathbf{b}^g$ error in Fig. 4, the results of our 2-frame and our 3-frame methods are largely consistent, because $\mathbf{b}^g$ is inherently solvable even in the two-frame case. This leads to only marginal differences compared to our 3-frame estimates. In contrast, the SNEC 3-frame method tends to produce highly deviant results under high angular velocity (manifested as spikes in the figure), indicating unstable performance. For velocity and gravity, which are unsolvable in the two-frame formulation, our 2-frame method is prone to increased errors under high acceleration, as evident in the early outliers in Fig. 4. This is an expected phenomenon as we employ approximations based on the average velocity and preintegrations, which is discussed in Section IV-C. Besides, our 3-frame method produces relatively stable results without the significant spikes observed in SNEC 3-frame.

Figure 5 illustrates the error of our 2-frame and 3-frame initialization under a varying acceleration threshold a_{th} , where frames with acceleration exceeding a_{th} are discarded. The results show that the two methods perform similarly when a is below 1 m/s^2 . Once a exceeds 1 m/s^2 , their performance diverges: 2-frame initialization suffers from increasing error, whereas 3-frame initialization remains largely stable. This suggests that acceleration below 1 m/s^2 is essential for achieving reliable 2-frame approximate initialization.

Figure 6 shows how sensitive the 3-frame initialization method is to depth errors on the MH01 and V101 sequences, with depth values corrupted by Gaussian noise of standard deviation σ_d . The results indicate that V101 is far more sensitive to depth inaccuracies than MH01. This is due to its much smaller depth range (around 2–5 m), where even small

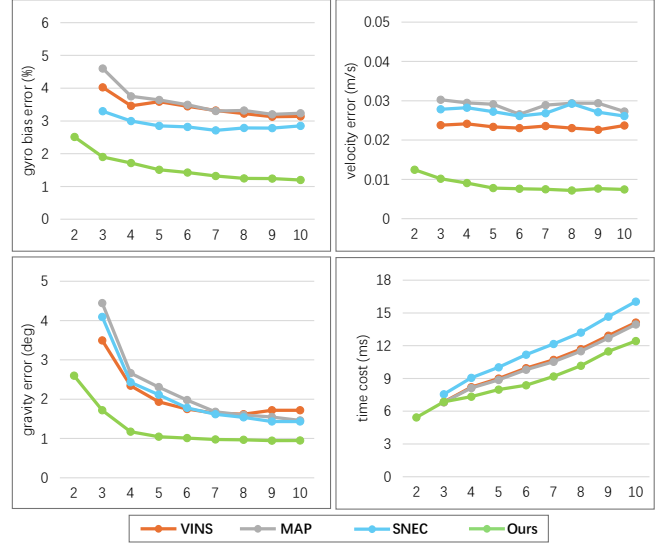


Fig. 7. Comparison of initialization methods across varying frame counts (ranging from 2 to 10). The results for the baselines (VINS, MAP, and SNEC) with two-frame initialization are omitted from the figure, as they cannot initialize successfully with only two frames.

TABLE III
TIME FOR INITIALIZATION WITH 10 FRAMES

	Stereo-Triangulation	Feature-Tracking	Visual-SfM	Initialization	Total
VINS	5.61	3.16	3.88	1.47	14.12
MAP	5.61	3.16	3.88	1.30	13.95
SNEC	5.61	3.16	3.88	3.39	16.04
Ours	5.61	-	-	6.81	12.42

depth variations lead to significant changes in the prediction. By contrast, MH01 operates at larger depths (6–10 m), where the influence of depth errors is considerably reduced.

D. Frame Counts and Time Evaluation

To further investigate the impact of the number of frames N on the performance, we evaluate all methods with frame counts ranging from 2 to 10. Figure 7 presents the results on the MH01 sequence, including the errors in gyroscope bias, velocity, and gravity, along with the runtime, across different initialization frame counts. Note the results for the baselines with two-frame initialization are omitted from the figure, as they cannot initialize successfully with only two frames.

As the number of frames increases, all methods demonstrate a general downward trend in $\mathbf{b}^g$ error and gravity error as shown in Fig. 7. Specifically, the $\mathbf{b}^g$ error of our method decreases from 2.5% when $N = 2$ to about 1%. In contrast, the figure for other methods decreases from around 4% to 3%, resulting in a final error approximately three times higher than ours. For velocity estimation, while N has little influence on accuracy, our method maintains errors around 0.01, while other methods consistently hover above 0.02. In the gravity estimation, the error of all methods decreases noticeably prior to five frames and largely plateaus thereafter. Our method achieves a final accuracy of around 1° , compared to 1.5° – 2°

TABLE IV
ABLATION RESULTS FOR 3-FRAMES INITIALIZATION

	α	β	$\mathbf{b}^g$	Vel.	G.Dir.	RRE	RTE
V101	-	-	2.67	0.013	3.61	0.027	0.0022
	✓	-	2.29	0.012	3.34	0.025	0.0021
	-	✓	2.43	0.012	3.40	0.025	0.0022
	✓	✓	2.29	0.012	3.31	0.025	0.0021
V102	-	-	5.57	0.027	3.26	0.038	0.0029
	✓	-	4.76	0.024	2.45	0.035	0.0026
	-	✓	5.03	0.025	2.57	0.035	0.0026
	✓	✓	4.72	0.024	2.44	0.034	0.0025
V103	-	-	13.67	0.049	3.05	0.077	0.0044
	✓	-	11.97	0.045	3.02	0.071	0.0041
	-	✓	12.70	0.047	3.03	0.074	0.0043
	✓	✓	11.47	0.045	2.98	0.070	0.0040

for the baselines. Overall, even with 3 frames for initialization, our method outperforms the baselines using 10 frames in most metrics.

The time cost during initialization gradually increases with the number of frames for all methods. The cost time of VINS and MAP is similar, at around 14ms when $N = 10$, while SNEC requires about 16ms due to the additional overhead of applying epipolar constraints and recomputing poses. Table III shows the runtime of each module for all methods with $N = 10$. The baseline methods take around 7ms on feature tracking and visual SfM combined, which accounts for nearly half of the total initialization time. By contrast, our method achieves the fastest initialization (12.42 ms), since it skips those steps and initializes directly.

E. Ablation Experiments for α and β

We additionally conduct an ablation study to validate the effectiveness of the coefficients α and β , as presented in Table IV. Introducing α and β individually both reduce errors, with α demonstrating more pronounced effects than β . The optimal outcome is achieved when both α and β are considered. It is noteworthy that V102 and V103 sequences exhibit pronounced improvement compared to V101. This is due to the large illumination changes present in these two sequences, and introducing α and β can effectively mitigate the impact of such image changes on the initialization.

VI. CONCLUSION

In this work we propose a direct initialization method for stereo visual-inertial odometry. Specifically, we construct a prediction function to estimate point coordinates from the initial parameters, which establishes the relationship between these parameters and the image intensity. This formulation allows us to solve the initialization by directly optimizing the photometric errors. It bypasses the feature tracking and visual initialization processes required in the conventional methods, which can introduce additional errors. Additionally, an approximate solution for the two-frame initialization is proposed. Extensive experiments on the EuRoC and TUM-VI datasets demonstrate that our method achieves state-of-the-art performance in both the estimation accuracy and initialization success rate.

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